

Portfolio Question 5.2.P - Conformal Prediction

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Portfolio 5.2.P (v4.0.0)

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- ▶ We claimed that conformal prediction could guarantee **any** model would get calibrated probabilities!
 - ▶ We only claimed it for marginal coverage. What is meant by this? When is it useful or not?
 - ▶ What methods are available for conditional coverage?
 - ▶ What data do we need to make it work?

Split conformal and marginal coverage

- ▶ Fit $\hat{f}$ on training data. On n separate **calibration** points compute scores $s_i = |y_i - \hat{f}(x_i)|$, and let $\hat{q}$ be the $[(n+1)(1-\alpha)]$ -th smallest. Then $C(x) = [\hat{f}(x) - \hat{q}, \hat{f}(x) + \hat{q}]$ satisfies

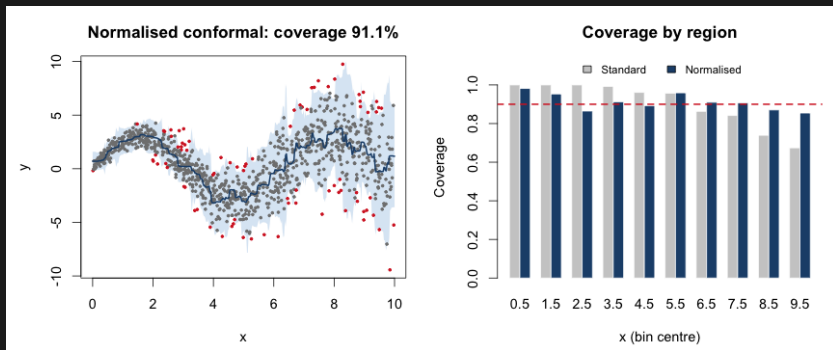
$$1 - \alpha \leq \Pr(Y_{new} \in C(X_{new})) \leq 1 - \alpha + \frac{1}{n+1}.$$

- ▶ **Why**: if the data are exchangeable, the new score's rank among the $n+1$ scores is uniform. Any model and any score function work.
- ▶ This is **marginal** coverage: the probability is over a random X_{new} **and** a random calibration set.
 - ▶ Useful for **aggregate** guarantees: 90% of all the intervals we issue will be right.
 - ▶ Not a guarantee for **this** x , or for a subgroup: some regions (or people) can be systematically under-covered.
- ▶ **Conditional** coverage, $\Pr(Y \in C(X) \mid X = x) \geq 1 - \alpha$ for all x , is **impossible** distribution-free with finite data, without infinitely wide intervals (Vovk 2012; Foygel Barber et al. 2021).

Towards conditional coverage

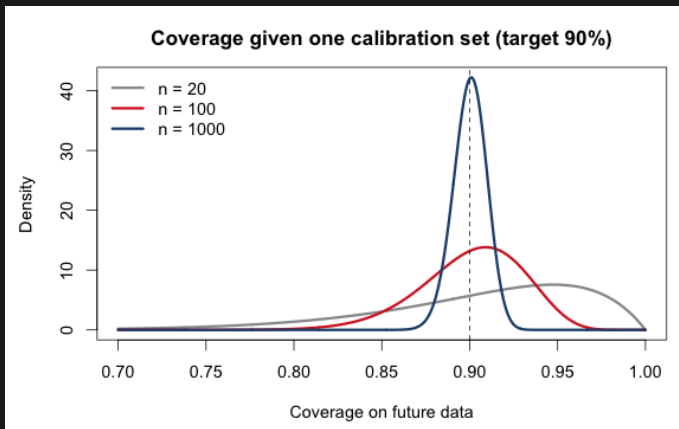
- ▶ We can't get conditional coverage for free, but we can get closer by choosing a better **score**:
 - ▶ **Normalised scores**: $s_i = |y_i - \hat{f}(x_i)| / \hat{\sigma}(x_i)$, with $\hat{\sigma}$ a model for the size of the error.
 - ▶ **Conformalised quantile regression** (CQR, Romano et al. 2019): fit quantile models $\hat{q}_{lo}, \hat{q}_{hi}$.
- ▶ Calibrate **within groups**: **Mondrian** conformal computes a separate $\hat{q}$ per group (class, cluster, etc), giving coverage **within each group** (e.g. for fairness).
- ▶ Every one of these keeps the marginal guarantee; a good model makes them *approximately* conditional.

Towards conditional coverage



- ▶ The same forest as the lecture, but with scores normalised by a second forest that predicts $|y - \hat{f}(x)|$.
- ▶ Coverage is now close to 90% everywhere, not just on average.

What data do we need?



- ▶ Averaged over calibration sets, the guarantee holds for any n .
Your calibration set may be unlucky.

Discussion

- ▶ There must be **Exchangeability** of calibration and future data.
When is this plausible?
- ▶ Would it work if we made clusters of size 1? How big should

Further reading

- ▶ **A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification** by Angelopoulos & Bates: start here. Sections 2 and 3 cover everything on these slides, with code.
- ▶ **Conformalized Quantile Regression** by Romano, Patterson & Candès (NeurIPS 2019).
- ▶ **The Limits of Distribution-Free Conditional Predictive Inference** by Foygel Barber, Candès, Ramdas & Tibshirani (Information and Inference, 2021).
- ▶ **Predictive Inference with the Jackknife+** by Barber, Candès, Ramdas & Tibshirani (Annals of Statistics, 2021).
- ▶ **Conformal Prediction Under Covariate Shift** by Tibshirani, Foygel Barber, Candès & Ramdas (NeurIPS 2019).
- ▶ Software: **MAPIE** (Python), and **conformalInference** (R, on GitHub).